Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/18/2025
PI Name:	Meng, Qingwu Ph.D.		
Project Title:	Temporal Lighting Optimization to Improve Lettuce Productivity and Nutritional Quality Under Superelevated CO2 Stress		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19716	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	09/01/2023	**End Date:**	08/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2025 per NSSC information (Ed., 7/31/24). NOTE: End date changed to 08/31/2024 per D. Freeland/KSC. Original end date per NSSC information was 12/31/2023 (Ed., 10/5/23).		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC23K1427		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Growing fresh food in space benefits crewmembers on long-duration missions by supporting nutrition and psychological wellbeing. Light-emitting diodes (LEDs) regulate plant photosynthesis and secondary metabolism, but fixed lighting protocols may not optimize growth or quality across crop development. This project evaluates temporal lighting strategies across crop developmental stages to improve growth, nutritional quality, and light use efficiency of red-leaf lettuce. Plants are grown hydroponically under LEDs with varying spectra (blue, green, red, and far-red) and intensity over time in controlled-environment chambers, under elevated or ambient-Earth CO_2. Data on biomass, morphology, pigmentation, and phytonutrient accumulation are collected to identify optimal light conditions for each growth phase. Results will guide crop-specific lighting strategies for efficient, high-quality plant production in space life support systems, supporting both spaceflight applications and controlled-environment crop production systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Temporal adjustments of light quality and intensity across crop growth phases enable precise photon delivery to optimize both yield and nutritional quality while improving light use efficiency. This project has identified spectral and intensity strategies that enhance lettuce growth and phytonutrient content efficiently, applicable to both spaceflight conditions and Earth-based controlled-environment agriculture, including indoor vertical farms. By optimizing photon allocation, these strategies support the efficient production of fast-growing, nutrient-rich crops for fresh consumption, providing a model for high-efficiency, high-quality food production in space and addressing the growing demand for fresh produce in urban and peri-urban environments on Earth.
Task Progress:	This project developed temporal lighting strategies to improve the growth, nutritional quality, and light use efficiency of red-leaf lettuce for long-duration space missions. Fresh food production in space provides essential nutritional and psychological benefits to crewmembers, and optimizing crop growth under controlled conditions is critical for efficient use of limited energy and space. Light-emitting diodes (LEDs) regulate plant photosynthesis and development, yet most current protocols use static light spectra and intensities throughout the crop cycle. Evidence suggests that plant responses to light vary with developmental stage, motivating the exploration of dynamic lighting approaches. Two fully replicated, controlled-environment experiments were conducted using hydroponically grown red-leaf lettuce. The first evaluated temporal changes in light spectrum (quality), while the second assessed temporal changes in light intensity. Experiments were performed under both elevated CO_2 and ambient-Earth CO_2 conditions to reflect realistic spacecraft scenarios. Across both studies, the effects of temporal light adjustments were assessed in terms of plant growth, morphology, pigmentation (including nutritional value), and overall biomass accumulation. Results indicated that temporal light adjustments can modulate plant development and resource use. Alternating light quality influenced leaf pigmentation and growth patterns depending on the timing and sequence of treatments. Temporal changes in light intensity allowed efficient photon delivery, supporting comparable plant growth to constant high light while reducing total energy input. Overall, dynamic lighting strategies demonstrated the potential to optimize lettuce production by balancing yield, quality, and energy efficiency, providing practical insights for space life support systems. The project provided significant training and professional development. Undergraduate and graduate students gained hands-on experience in experimental design, hydroponic crop production, data collection, statistical analysis, and scientific communication. Two undergraduates conducted independent experiments, interpreted results, and presented findings at the 2025 American Society for Horticultural Science conference, earning awards in the Undergraduate Poster Competition. Other trainees contributed to technical support and experimental execution under the mentorship of the principal investigator, acquiring skills in controlled environment agriculture and research methodology. Outreach and dissemination included professional conference presentations and local university events, highlighting the relevance of temporal lighting strategies for both space-based crop production and efficient indoor agriculture on Earth. Data generated have informed manuscripts currently under preparation or review. Overall, the project successfully demonstrated that temporal modulation of light quality and intensity can enhance lettuce growth efficiency and nutritional potential. These strategies provide guidance for crop-specific lighting protocols in controlled environments, contributing to the advancement of space agriculture and efficient, high-quality indoor food production on Earth.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Ding S, Meng Q. "Temporally increasing light intensity produces similar lettuce growth more efficiently than fixed high light." 2025 American Society for Horticultural Science (ASHS) conference, New Orleans, Louisiana, July 28-August 1, 2025. Abstracts. 2025 American Society for Horticultural Science (ASHS) conference, New Orleans, Louisiana, July 28-August 1, 2025. HortScience, 60(7S):S1-S496. https://doi.org/10.21273/HORTSCI.60.7S.S1 , Jul-2025
Abstracts for Journals and Proceedings	McCann T, Meng Q. "Temporal light quality or intensity alternations elicit sequence-dependent intermediate lettuce growth compared to fixed lighting." 2025 American Society for Horticultural Science (ASHS) conference, New Orleans, Louisiana, July 28-August 1, 2025. Abstracts. 2025 American Society for Horticultural Science (ASHS) conference, New Orleans, Louisiana, July 28-August 1, 2025. HortScience, 60(7S):S1-S496. https://doi.org/10.21273/HORTSCI.60.7S.S1 , Jul-2025
Articles in Peer-reviewed Journals	Kang I, Ding S, Meng Q. "Temporally increasing light intensity produces similar lettuce growth more efficiently than fixed high light." Front Plant Sci. 2026 Feb 3;17:1763702. https://doi.org/10.3389/fpls.2026.1763702 , Feb-2026